From Price to Profit: Multi-Tier Private Label Response Across Retail Formats *

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Abstract

This study examines how pricing affects sales and profitability of multi-tier private labels (PLs) across store formats within a single retail chain. Using scanner data from a Nordic grocery retailer operating hypermarkets, supermarkets, and convenience stores, we estimate error correction models linking price changes to sales and profits for budget, standard, and premium PL tiers. Results show that own-price elasticities are negative for standard and premium tiers, while the budget tier exhibits heterogeneous responses. Cross-tier substitution effects are limited, indicating that PL tiers largely function independently. Profit elasticities are positive across tiers, with format-specific differences: hypermarkets gain most from budget-tier pricing, while supermarkets benefit from standard and premium tiers. The findings demonstrate that pricing effectiveness is tier- and format-specific, highlighting the need for more format-contingent pricing strategies, where the role of each PL tier in pricing decisions varies across retail formats rather than following a uniform approach.

Keywords: Grocery retailing; Private labels; Pricing; Store formats

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1 Introduction

In recent years, a grocery retail landscape has seen a significant shift towards the adoption of private labels (PLs), store brands owned by the retailer, reflecting broader consumer acceptance and retailer strategies. Growing from 31% in 2018 to 38% in 2024, private labels have claimed a significant share of the grocery retail market value in Europe (IRI, 2024). The popularity of private labels is evident not only in hard discount stores but also in traditional supermarkets, illustrating their common appeal across different retail formats (GfK, 2021). Private labels are usually diversified into various tiers such as budget, standard, and premium tiers (Keller et al., 2022). Each tier is tailored to cater to distinct consumer segments and needs. With the growing popularity of private labels, many grocery retail chains tend to offer a tiered assortment across all their stores within a chain. Yet, managing private labels across different store formats is challenging since each format serves different shopping missions and faces operational constraints, such as limited shelf space or assortment depth (e.g. Gauri et al., 2008). Thus, retailers must adapt their multi-tier PL strategies by adjusting pricing and tier emphasis to match the shopping behavior and constraints of each store format. In other words, rather than applying a uniform pricing strategy across all formats, retailers may benefit from placing different pricing emphasis on specific PL tiers depending on the store format. This does not necessarily imply fully localized store-level price differences, but rather format-contingent pricing decisions within a unified pricing system.

Prior research suggests that retailers often maintain relatively uniform prices across stores, despite variation in local demand and competition. This tendency has been attributed to factors such as brand image concerns, consumer perceptions of fairness, and organizational constraints (DellaVigna & Gentzkow, 2019). Moreover, while local price differentiation can improve profitability under certain conditions, uniform pricing may also be optimal when it helps soften competition across markets (Li et al., 2018). As a result, retailers may not fully adjust prices across stores. This makes it important to understand how pricing effectiveness varies across formats and tiers within existing pricing structures. This is particularly relevant for private labels, where retailers have greater control over pricing and can adjust tier-level strategies to align with format-specific objectives.

To design effective pricing strategies for multi-tier PLs across formats, retailers require empirical insights into how price changes affect sales and profitability within each format. While private label tiers are often deployed uniformly across store types, their actual performance may differ. Previous research has extensively examined the impact of introducing different tiers of PLs on brand, category, and store performance (e.g., Geyskens et al., 2010; Gielens, 2012; ter Braak et al., 2013), and how these tiers perform across different types of retail chains (i.e., supermarkets versus discounters) (e.g. Hökelekli et al., 2017). However, there has been limited focus on how multi-tier PLs perform across different store formats within the same retail chain, particularly in terms of their sales–profitability relationship. Prior research tends to examine sales or profitability in isolation, without tracing the chain of effects from price changes to sales to profit.

To address these gaps, this study investigates how multi-tier PLs respond to pricing strategies across physical store formats within the same retailer. Specifically, we examine whether differences in price effectiveness on sales persist across formats and how these differences translate into format-specific profitability. In doing so, we focus on variation in observed prices over time within each store format, rather than comparing static price differences across formats. To do so, we propose a framework that links price changes to sales and then to profit, accounting for both own-tier and cross-tier effects. This framework allows us to isolate substitution patterns among PL tiers and assess whether they function independently, complement one another, or cannibalize sales across tiers. By doing so, we examine not only the responsiveness of each tier to price changes but also the degree to which PL tiers collectively enhance or reduce overall profitability within each store format. This question is not only empirically practical but also responds to the call for future research opportunities proposed by Keller et al. (2022) on understanding retailers' overall profitability with multi-tiered PLs. Understanding how pricing impacts tier-level performance across different store formats provides academic insights into private label portfolio literature and offers actionable guidance for retailers seeking to enhance store-level profitability.

Thus, the purpose of this study is answer the question by empirically examining the effects of pricing on the sales and profits of each tier of private labels across various store formats within the same retail chain. We utilize detailed scanner data available from a Nordic grocery retailer, which operates under three specific store formats including hypermarkets, supermarkets, and convenience stores and offers multi-tier private labels categorized into the budget tier, the standard tier and the premium tier. Our analysis is based on observed transaction prices, which reflect the effective prices paid by consumers and incorporate both regular and promotional pricing. We explore how price adjustments in one tier may influence sales in others, capturing cross-tier responsiveness and identifying whether interactions among tiers reflect substitution, complementarity, or independence. Specifically, we first estimate how changes in the prices of each PL tier affect their respective sales within the same format. We then use these estimated changes in sales to assess how price adjustments consequently impact overall store profitability, isolating the contribution of each tier while accounting for interdependencies across tiers. This step-by-step approach allows us to clearly delineate the direct effects of price changes on sales volume and their subsequent effects on profitability.

Our findings suggest that the performance of multi-tier PLs varies significantly across store formats, both in terms of sales responsiveness and profitability. While standard and premium PL tiers consistently exhibit negative price elasticity, the magnitude of this effect differs across formats, with the hypermarket and the supermarket showing distinct responsiveness. Budget tiers display more heterogeneous patterns, including instances of positive price elasticity, implying varying perceptions of value. Moreover, we find limited evidence of cross-tier substitution, particularly for the premium tier, which shows minimal responsiveness to price changes in lower tiers across most categories and store formats. This suggests that consumers purchasing premium PLs tend to view them as distinct offerings rather than interchangeable with budget or standard alternatives, supporting prior findings that premium products often serve niche or quality-focused segments with weaker substitution tendencies. As a result, pricing decisions within the premium tier may have less impact on broader portfolio dynamics, reinforcing the need for tier-specific strategies. These differences in sales responses translate into divergent profitability outcomes; the hypermarket tends to benefit more from pricing strategies targeting the budget tier, while the supermarket see stronger profit returns from standard and premium tiers. The convenience store exhibits limited pricing variation and restricted cross-tier interactions, reflecting its narrower assortment and targeted format strategy.

In summary, our study demonstrates that the effects of pricing on multi-tier PLs are highly format- and tier-specific. By tracing the link from price changes to both sales and profitability, we show that budget, standard, and premium PLs respond differently across hypermarket, supermarket, and the convenience store. Limited cross-tier price responsiveness suggests that most tiers function independently, enabling retailers to manage each tier with minimal concern for cannibalization. However, in cases where pricing in one tier influences another, particularly between the budget and standard tiers, retailers should carefully evaluate the profitability implications. Our step-by-step approach offers a practical way to quantify how price changes in any given tier impact total store profitability, supporting more informed pricing decisions across the private label portfolio.

The rest of the paper is organized as follows. We provide a brief overview of relevant literature and our research framework that illustrates the relationships among profit, price, cost, and quantity of sales across multi-tier private labels in Section 2. We introduce the empirical setting, data and measure for this study in section 3. We specify our step-by-step approach to delineate the direct effects of price changes on sales volume and their subsequent effects on profit margins in section 4. We present and discuss our results in Section 5 and 6 respectively.

2 Background and Research Framework

2.1 Literature Review

The previous literature on private labels extensively covers their sales performance, profitability, and market segmentation strategies (Table 1). Yet, few studies examine how pricing influences both sales and profitability across multiple PL tiers and store formats within the same retail chain. In particular, the sequential link between price changes, sales volumes, and resulting profits across tiers and formats is rarely addressed altogether. Prior research demonstrates how PLs and their multi-tier structure can contribute to retailers. Pauwels & Srinivasan (2004) found that the entry of PLs benefits retailers through higher unit margins but does not necessarily drive store traffic nor revenue growth. However, at the customer level, heavy PL users contribute less to total retailer profit than light PL users due to their tendency to purchase lower-priced items (Ailawadi & Harlam,2004). Moreover, budget[1] PLs tend to reduce retailer margins and cannibalize standard PLs, while increasing standard NB sales and overall category sales (Geyskens et al., 2010; Gielens, 2012). In contrast, standard and premium PLs are generally more effective at driving category growth than budget PLs.

Table 1: Selected literature on PLs performance in sales and profits

Paper	**Sales**	**Profits**	**PL Comparison**	**Format Comparison**
Paulwels & Srinivasan (2004)	Yes	Yes	PLs vs NBs	No
Ailawadi & Harlam (2004)	Yes	Yes	PLs vs NBs	No
Ailawadi, Paulwels & Steenkamp (2008)	Yes	No	PLs vs NBs	Service chain vs Value chain
Geyskens, Gielens & Gijsbrechts(2010)	Yes	No	Budget vs Standard vs Premium	No
Gielens (2012)	Yes	No	Budget vs Standard vs Premium	No
ter Braak, Dekimpe & Geyskens (2013)	No	Yes	Budget vs Standard vs Premium	No
Sethuraman & Gielens (2014)	Yes	No	PLs vs NBs	No
Keller, Dekimpe & Geyskens (2016)	Yes	No	PLs vs NBs	No
Hökelek, Lamey & Verboven (2017)	Yes	No	Budget vs Standard vs Premium	Traditional retailers vs Discounters
Widdecke, Keller & Gedenk (2023)	Yes	No	PLs vs NBs	Supermarkets vs Discounters
van der Plas, Dekimpe & Geyskens (2024)	No	Yes	PLs vs NBs	Conventional retailers vs Hard discounters
This Study	**Yes**	**Yes**	**Budget vs Standard vs Premium**	**Hypermarket vs Supermarket vs Convenience Store**

Different retail formats can moderate the role of PLs. Ailawadi, Pauwels & Steenkamp (2008) and Widdecke et al. (2022) found that found that service-oriented retailers tend to foster higher PL loyalty, whereas discounters benefit more from aggressive PL pricing strategies to attract price-sensitive consumers. Hökelekli, Lamey & Verboven (2017) found that standard PLs are the most effective in helping traditional retailers compete with discounters, while economy PLs tend to cannibalize other PL tiers. Van der Plas, Dekimpe & Geyskens (2024) demonstrated that

[1] Following the terminology used in Keller et al. (2022), this study refers to 'economy' private labels as 'budget' private labels.

consumers at hard discounters are highly responsive to price differences between national brands and private labels, both within the same store and across store formats.

While these studies offer valuable insights, existing research has primarily focused on the competition between private labels and national brands with limited finding regarding internal competition within retailers' multi-tier PL portfolios. Retailers increasingly invested in their multi-tier PLs not simply to counter national brands, but to strategically segment customers, optimize margins, and differentiate their store image (Nielsen, 2025). Understanding of how pricing affects each tier's performance, both individually and in relation to other tiers, can help retailers mitigate the risk of undermining profitability through unintended substitution effects or inefficient pricing structures. Unlike competition with national brands, intra-PL strategies are under the retailer's direct control, offering flexibility and potential for sustained competitive advantage.

Building upon these findings, our study contributes to the literature by examining the performance of PL tiers across different store formats within the same retail chain. Unlike prior research that primarily emphasizes PL–NB competition or focuses on aggregate format comparisons, we investigate how intra-PL pricing dynamics, both within and across tiers, influence sales and profitability at the format level. In response to the research opportunities highlighted by Keller et al. (2022), we explicitly investigate the sequential relationship between price changes, sales volumes, and resulting profits. This approach not only addresses the call for a more detailed investigation into the impact of multi-tier PLs across varied retail environments but also provides actionable implications for tailoring PL strategies by format to enhance total retailer performance.

2.2 Research Framework

Building on the identified theoretical and empirical research gaps, we propose the following research question. Given the differences in price effectiveness on sales among multi-tier PLs, do these differences persist across different store formats, and could this lead to varying levels of profitability across store formats? To do so, we follow our framework (Figure 1) that illustrates the relationships among profit, price, cost, and quantity of sales across multi-tier private labels.

The change in profit for each private label tier (budget, standard, premium) can be driven by changes in price, cost, and the corresponding changes in quantity sold. This relationship is captured as Δ Profit (π) where $\Delta\pi = f(\Delta Price, \Delta Cost, \Delta Sales)$[2]. Our framework proposes that the changes in quantity sales for each tier (Δ S: Budget, Δ S: Standard, Δ S: Premium) can be influenced by changes in price across all tiers (Δ P: Budget, Δ P: Standard, Δ P: Premium), suggesting an interdependent price dynamic among the tiers. In other words, we assume that the price of one tier can potentially affect the sales volumes of another, addressing interactions within a multi-tier pricing strategy documented by previous studies. By focusing on ΔS, we aim

[2]The full equation representing the change in profit can be approximated and expressed as: $\Delta\pi = S \cdot (\Delta P - \Delta C) + \Delta S \cdot (P_t - C_t)$

Δ Profit = f (Δ Price (P), Δ Cost (C), Δ Sales(S))				Δ P Budget	Δ P Standard	Δ P Premium
Δ π	Δ C	Δ P	Δ S	Δ S = f(P)		
Δ π : Budget	Δ C : Budget	Δ P : Budget	Δ S : Budget	Budget Brand Price Elasticity	Cross Price Elasticity	Cross Price Elasticity
Δ π : Standard	Δ C : Standard	Δ P : Standard	Δ S : Standard	Cross Price Elasticity	Standard Brand Price Elasticity	Cross Price Elasticity
Δ π : Premium	Δ C : Premium	Δ P : Premium	Δ S : Premium	Cross Price Elasticity	Cross Price Elasticity	Premium Brand Price Elasticity

- Brand Price Elasticity
- PL-tiers Cross Price Elasticity (Downward Substitution)
- PL-tiers Cross Price Elasticity (Upward Substitution)

Figure 1: Multi-Tier Private Label Profitability: (Cross) Price Elasticity and Substitution Effects Framework

to directly capture the responsiveness of sales volumes of each private label tier to price changes.

We utilize this framework to establish an empirical model that quantifies the impact of price changes on profits across different store formats. This model will incorporate the inter-tier price effects to more accurately predict profitability shifts in response to pricing strategies.

Despite proposing a structural relationship between price, sales, and profit across PL tiers, our framework does not hypothesize specific expected relationships between cross-tier price changes and sales. The complexity of market dynamics and consumer behavior allows for a range of possibilities, including upward substitution, where changes in the prices of Budget or Standard tiers might influence the demand for higher-tier brands, and downward substitution, where changes in the prices of Standard or Premium tiers could affect demand for lower-tier brands (Figure 1). The relationship of price changes across tiers (i.e., upward substitution and downward substitution) may result in positive, negative, or null effects on sales volumes. The null effects occur when each tier has significant different positioning with different customer segments (Martos-Partal et al., 2013).

Moreover, our framework does not impose specific budget constraints within categories, allowing us to propose a broader range of consumer responses to price changes without restricting the scope of potential impacts. The absence of predefined budget constraints acknowledges that consumer purchasing decisions are not solely driven by category-specific budgets but are also influenced by perceived value, brand loyalty, and available alternatives across the entire store offering.

If PL tiers can be substitutable by one another, the relationship of price changes across tiers will be positive meaning that an increase in one tier price could lead to an increase in another tier price by making another tier become relatively cheaper. For example, if the price of the Premium tier increases, consumers may find the Standard or Budget tiers more appealing, not just because of a lower price, but also due to the diminished perceived value gap between the tiers. When faced with a set of alternatives, a relative price increase in one option can significantly enhance the attractiveness of the others, provided the goods are considered close substitutes (e.g., Lichtenstein

& Bearden, 1989). Conversely, if the price of the Budget tier were to increase, this could narrow the perceived price gap between the Budget and higher tiers such as Standard and Premium. Under these circumstances, consumers might reassess their value-for-money proposition. For instance, a smaller price differential might make the Standard and Premium tiers appear more attractive, offering perceived better quality or features for a slight increase in cost (e.g., Rao & Monroe, 1989). This could encourage consumers to "trade up" within the same brand's product line, especially if the quality differential is perceived to justify the relatively small additional expense.

The relationship of price changes across tiers can also be negative. This means that an increase in the price of one tier might lead to a decrease in the sales of another. For example, if the price of the Premium tier increases significantly, it might lead to a decrease in the sales of the Standard and Budget tiers. This can occur because the higher price point of the Premium tier may elevate the overall brand perception to a premium level, causing the brand to be perceived as less accessible or too expensive for budget-conscious consumers. This can alienate a portion of the brand's customer base who previously might have considered products from the Standard or Budget tiers but now feel that the brand on the whole has moved beyond their typical spending range (e.g., John et al., 1998; Kahneman et al., 1986). An increase in the Premium tier's price might also set new price expectations for the brand, making the Standard and Budget products seem less appealing to consumers who do not perceive enough quality or value difference to justify the cost, even at lower price points (e.g., Lichtenstein et al., 1993; Ailawadi et al., 2001). As a result, customers might opt for competing brands where they perceive better value across the price spectrum, resulting in an overall decline in sales across the Standard and Budget tiers. On the other hand, an increase in the price of the Budget tier could also negatively impact the sales of the Standard and Premium tiers. This scenario can occur through a process known as the negative halo effect, where the increased price of the Budget products may lead consumers to perceive the entire brand as less affordable or less value-driven, thus deterring these consumers from engaging with the brand altogether, rather than encouraging them to trade up to higher-priced tiers (e.g., Zeithaml, 1988). This could result in a significant segment of the customer base leaving for more competitively priced alternatives, thereby reducing the overall traffic and potential sales across all tiers. The price increase at the Budget level might also reset consumer expectations regarding the brand's pricing strategy, making even the Standard and Premium tiers seem less attractive if consumers believe that the brand is generally moving towards higher price points (e.g., Hamilton & Chernev, 2013). This shift in perception could lead to decreased sales in the higher tiers as consumers reevaluate their willingness to pay within the context of the new pricing structure, potentially seeking better-perceived value elsewhere.

In summary, this research leverages the outlined framework to construct an empirical model that systematically investigates the interconnected relationships between price, sales, and profit across different private label tiers within the same retail chain. Drawing on established findings from previous literature and theories in consumer behavior, we formulate the initial structure of our model to reflect these scholarly insights. This approach ensures that our model is theoretically sound. By intentionally not imposing any initial hypotheses on the relationships of cross-price elasticities, we maintain the flexibility to empirically uncover whether the influences are positive, negative, or null. This allows us to adapt our expectations based on observed data, accommodating a range of outcomes that reflect the variety of consumer responses to price changes.

3 Data and Measures

3.1 Empirical Setting

To investigate whether the differences in price effectiveness among multi-tier PLs persist across various store formats and how this influences profitability levels within those formats, we utilize detailed scanner data available from January 1, 2014, to November 30, 2016 obtained from a Nordic grocery retailer. The retailer operates under three specific store formats including hypermarkets, supermarkets, and convenience stores. Each format is designed to meet the varied shopping needs and preferences of different customer segments across regions. Moreover, the retailer owns the private labels which are accounted for more than 20 % of sales across these regions.

The retailer operates several private labels, which we have categorized into three tiers based on their distinct customer appeal and product description. These tiers follow the retailer's internal classification, which reflects consistent positioning in terms of price, quality, and target segment across store formats. The budget tier offers everyday items at competitive prices through large-scale purchasing and simple packaging, tailored to meet the needs of consumers who prioritize price over premium features. The standard tier offers a diverse product range, from basic staples sourced from local manufacturers to international producers, all noted for their consistent quality and often associated with sustainable benefits. The premium tier features ecologically produced goods and is marketed as high-quality products with a focus on distinctive taste, and regional authenticity. These premium products are typically promoted as suitable for special occasions or customers seeking an elevated shopping experience, reflecting how the retailer differentiates them from lower-tier alternatives.

In contrast, national brands are not organized into clearly defined tiers within the data. While price differences exist across national brands, these do not map cleanly into consistent tier categories due to variation in brand positioning and product attributes across categories. For this reason, we do not impose a tier structure on national brands based solely on price levels.

3.2 Data

3.2.1 Selected Stores and Relevant Products

The dataset is obtained from the retailer's loyalty membership database. The data includes detailed purchase information from selected customers, whose initial purchase is from the hypermarket. Their purchase information and shopping behavior information were then tracked across all store formats within the same retailer chain. Since these customers can make purchases at any store within the same retail chain, we minimize potential geographic and economic disparities by zooming in on three specific stores, representing one of the formats (hypermarket, supermarket, and convenience store) and located within the same city and most frequently visited by these customers.

Although our dataset provides detailed customer purchase information across store formats, it limits information to only the brands and categories available in our loyalty data. To overcome this, we focus on analyzing multitier PLs that are consistently purchased across the selected hypermarket, supermarket and convenience store. To ensure comparability across formats, we further restrict the analysis to a consistent set of products observed in multiple store formats, thereby minimizing differences driven by the product mix. This allows us to more accurately assess pricing strategies and brand performance across these formats.

First, we select PL products that are purchased at least once every two weeks, ensuring sufficient variability in price and performance observations over the data collection period. This threshold helps us avoid analyzing products with sporadic or seasonal sales patterns that could skew elasticity or profitability estimates. Next, we include only those national brand products that share similar descriptions with our selected PLs, guaranteeing that our comparisons are made between equivalent items. Finally, because our aim is to compare both within-brand tiers and across store formats, we limit our scope to product categories that have at least two multi-tier PLs offered in at least two different store formats. This restrictive selection helps maintain the validity of our comparisons across tiers and formats.

To illustrate this selection process, consider the category of biscuits. Although biscuits are available across different PL tiers and several national brands, not all product variants are suitable for comparison. Some biscuit types, such as gingerbread, are highly seasonal and typically purchased only during specific holidays like Christmas. These items do not meet our biweekly purchase threshold and could distort estimates of price elasticity and profitability. Others, such as whole grain or specialty biscuits, may be exclusive to premium PLs or high-end national brands, making them unavailable in lower-tier PLs or mass-market brands. Including such products would compromise comparability across both tiers and store formats. Therefore, in line with our selection criteria, we focus on commonly available product descriptions, such as plain digestive biscuits, that are consistently offered under all PL tiers and by major national brands, and are present in at least two different store formats. This ensures that the products we analyze serve similar consumer needs and enables us to isolate the effects of tier and format without confounding variation from product differentiation or seasonal demand.

After selecting, we retain five categories including biscuit, canned vegetable, cereals, pasta and raw nuts and dried fruits (Table 2). The availability of these products across different store formats varies. While the hypermarket and supermarket generally offer a wider range of products across all three brand tiers (basic, standard, premium), the convenience store has a more limited selection, typically featuring only basic and standard tiers, with only one category available. This difference likely stems from the space constraints and targeted assortment strategies characteristic across physical store formats. It is worth noting that our empirical setting competition between the basic and standard tiers, with no evident instances of competition between standard and premium tiers.

Table 2: Selected Multitier (B: Budget, S: Standard P: Premium) PLs Products and Their Availability Across Different Formats

Category	**Format**	**Example of product description**	**B**	**S**	**P**
Biscuit	Hypermarket	digestive	x	x	
	Supermarket		x	x	
Canned Vegetables	Hypermarket	tomato, corn, champignon, olive	x	x	x
	Supermarket		x	x	x
	Convenience		x	x	
Cereals	Hypermarket	cornflake, muesli, raisin	x	x	x
	Supermarket		x	x	
Pasta	Hypermarket	spaghetti, penne, gnocci	x	x	x
	Supermarket		x	x	x
Raw nuts, dried fruits	Hypermarket	plum, raisin, apricot	x	x	x
	Supermarket		x	x	x

Despite the wide assortment of products offered by the retailer, our final selection is limited to five product categories. This limited number of categories results from both data constraints and the stringent criteria necessary for valid cross-format comparisons. First, only a subset of categories features products available across PL tiers (budget, standard, premium) and across at least two store formats. Many categories either lack a full PL tier structure or are not consistently present across store formats particularly in the convenience store. Second, to support reliable estimation, we retain only products that meet our earlier biweekly purchase threshold.

Consequently, only one category, which is canned vegetables, meets these criteria across all three store formats, including the convenience store. While this limited representation might appear restrictive, it remains analytically meaningful. First, we define our research scope and selection criteria ex-ante, independent of final product availability, to maintain transparency and avoid selection bias. Second, the inclusion of the convenience store, even with just one qualifying category, provides a valuable contrast. It allows us to explore how multi-tier PLs perform in the most space-constrained and operationally distinct retail format. Rather than generalizing across all categories, our findings from the convenience store offer format-specific insights, highlighting the dynamics and limitations of PL tier strategies in constrained environments where assortment depth is minimal.

We summarize our final data on customer spending across selected product categories by store format in Table 3. The summary of customers spending across Multitier PLs implies varied multitier PLs performances across formats. The hypermarket, the largest format, has the most visitors (4,482) but shows moderate market share for budget (11%) and premium (10%) PL tiers, with relatively large share on standard tier products (38%). The supermarket, though smaller, has higher market share in the premium tier (15%) compared to the hypermarket. It is important to note that in our study, the convenience store is analyzed with only the canned vegetable category, yet it shows higher market shares for both the budget (36%) and standard (48%) tiers. Despite the varying popularity between the budget and premium tiers across five categories, the standard tier appears as the most established and dominant PL tier in our setting.

Table 3: Customers and their spending on selected product categories by store format

	Hypermarket	**Supermarket**	**Convenience**
Size (m^2)	4,252	2,155	497
Total unique brands available	39	40	13
Total unique products available	306	282	62
Total visitors	4,428	2,927	915
Avg Market Share for Budget	11	7	36
Avg Market Share for Standard	38	28	48
Avg Market Share for Premium	10	15	-

Notes: Reported across 135,776 shopping trips of five selected product categories (though, one for the convenience store) from 4,921 customers from the same demographic area visiting these store formats. The table reported the size (square meters), the number of brands and products observed, and the average market share for each PL tier in each format per visit (%).

3.3 Variable Operationalization

Table 4 provides details about the operationalization of the variables in our study. We focus primarily on two dependent variables: sales and profits. The main independent variable in the empirical analysis is the price of PLs, which is constructed as a sales-weighted average across its SKUs (Datta et al., 2022).

In addition to PL prices, we include several control variables to account for broader market conditions and demand fluctuations. The national brand price is operationalized as the market share–weighted average price of all other brands within the same category, excluding the focal private label tiers (budget, standard, and premium), and serves as a control for the competitive pricing environment. We further control for promotional intensity using feature and display activity, as well as seasonal effects through holiday indicators.

All variables are aggregated biweekly (referred to as *week t* throughout the analysis) to ensure continuity in our data and to prevent loss of information from excessive granularity. This aggregation allows for appropriate temporal adjustments that can capture lagged effects between price adjustments and brand performance. The final panel consists of 76 biweekly periods for each category–tier–format combination.

Table 4: Variable Operationalization

Variable	**Description and Operationalization**
$S_{ijk,t}$	Total Unit sales for PL Tier i in category j in store format k in week t.
$\text{Profits}_{ijk,t}$	Total Profits for PL Tier i in category j in store format k in week t.
$\text{Price}_{ijk,t}$	Price of PL Tier i in category j in store format k in week t, computed as a sales-weighted average across its SKUs sold in period t, expressed in local currency per unit (i.e., 100 gram).
$\text{NBPrice}_{jk,t}$	Average price of national brands in category j in store format k in week t, computed as a sales-weighted average across their SKUs sold in period t, expressed in local currency.
$\text{cost}_{ijk,t}$	Costs of PL Tier i in category j in store format k in week t, computed as a sales-weighted average across its SKUs sold in period t, expressed in local currency.
$\text{FD}_{ijk,t}$	Intensity of feature and/or display support for PL Tier i in category j in store format k in week t, computed as a sales-weighted average across its SKUs sold in period t, expressed in number range from 0 to 1.
Holiday_t	An indication variable equal to 1 it is a national holiday in week t.

3.4 Descriptives

Table 5 and 6 present descriptive statistics for multitier PLs prices and performances across different store formats within our dataset. We observe that the standard tier consistently dominates in terms of sales across all categories and store formats. However, the performance between budget and premium tiers varies; for instance, in the hypermarket within the cereals category, sales for budget and premium tiers are nearly equal, whereas in the pasta category, such gaps are less evident in the supermarket. Despite these variations in sales, total profits across tiers generally align with tier positioning, with higher tiers typically yielding higher profits (Keller et al., 2022).

Regarding PLs pricing, the budget tier consistently offers the lowest prices, but the gap between the standard and premium tiers is not uniform across categories. For example, in the category of canned vegetables, the price of standard tier products is closer to those of the premium tier, indicating a smaller price differentiation. In contrast, in the pasta category, the price gap places the standard tier closer to the budget tier, suggesting a broader pricing strategy that varies significantly by category.

In addition to cross-sectional differences, prices seem to vary over time within each store format, providing the empirical variation used in our analysis. Price adjustments occur regularly across categories, although they are not always implemented simultaneously across store formats, reflecting format-specific pricing decisions. It also shows that prices for the same PL tiers are not identical across formats, indicating that retailers already apply some degree of format-specific pricing. These differences are evident across several categories, where identical tiers are priced at different levels in the hypermarket, the supermarket, and the convenience store.

Importantly, these observed price differences are not driven by aggregation or assortment composition effects, as our analysis is conducted at the product level using consistent brand–product definitions across formats. This ensures that the estimated price effects reflect actual price variation rather than differences in the assortment mix.

4 Method

In the research framework, we propose that the change in profit for each private label tier can be influenced by changes in prices and corresponding changes in quantity sold for that tier while the changes in sales for each tier can be affected by changes in prices across all tiers. This structure addresses an interdependent price dynamic among the tiers and potential cannibalization effects within PLs. To assess how price adjustments consequently impact overall store profitability, taking into account the distinct contributions of each PL tier, we first estimate sales-response models for each tier (in each category and store format) and calculate estimated changes in sales for given changes in prices of other PL tiers. We then estimate profits-response models for each tier (in each category and store format) to obtain the effect of prices on profits. Table 7 summarizes the steps implemented in this study with their purposes and corresponding approach and relevant variables and parameters.

4.1 Sales-Response Model Specification

In order to quantify the effect of price changes on sales across different PL tiers, we adopt an error correction (EC) specification. This specification allows us to estimate the short-run elasticity of prices while accounting for potential long-run equilibrium relationships (Pauwels et al., 2007). In our sales response model specification, we use linear model to avoid the aggregation bias that might happen in another transformed specification such as log-log specification (Datta et al., 2022). Moreover, the linear model allows us to directly calculate estimated changes in sales without the need for further transformation.

Adopting an error-correction specification, the model for PL tier i sales in category j in store

Table 5: Descriptive statistics of multitier PLs performances across store formats

	Hypermarket	**Supermarket**	**Convenience**
Biscuit (unit: 100 gram)			
Budget PL Sales	53.47 (25.99)	31.89 (16.96)	-
Budget PL Profits	20.53 (12.55)	11.92 (7.39)	-
Standard PL Sales	429.32 (149.91)	71.11 (45.18)	-
Standard PL Profits	485.34 (133.13)	107.57 (47.55)	-
Canned Vegetables (unit: 100 gram)			
Budget PL Sales	1,333.95 (261.57)	351.99 (75.12)	142.90 (63.00)
Budget PL Profits	388.56 (103.71)	92.21 (26.54)	69.33 (22.77)
Standard PL Sales	3,502.93 (632.59)	836.66 (288.46)	134.57 (71.93)
Standard PL Profits	2,864.85 (416.07)	791.85 (147.00)	141.71 (43.29)
Premium PL Sales	382.84 (212.31)	166.16 (69.67)	-
Premium PL Profits	279.38 (92.01)	146.64 (63.13)	-
Cereals (unit: 100 gram)			
Budget PL Sales	168.22 (38.37)	33.22 (15.09)	-
Budget PL Profits	10.01 (7.06)	1.50 (2.85)	-
Standard PL Sales	565.24 (111.87)	203.64 (46.81)	-
Standard PL Profits	226.53 (63.49)	105.73 (26.80)	-
Premium PL Sales	119.54 (87.44)	-	-
Premium PL Profits	67.02 (34.61)	-	-
Pasta (unit: 100 gram)			
Budget PL Sales	522.37 (103.74)	64.34 (27.06)	-
Budget PL Profits	120.58 (27.86)	11.27 (4.99)	-
Standard PL Sales	383.95 (100.43)	116.64 (45.76)	-
Standard PL Profits	97.59 (44.96)	34.32 (13.50)	-
Premium PL Sales	76.12 (42.78)	33.16 (16.24)	-
Premium PL Profits	44.52 (24.48)	16.14 (8.94)	-
Raw nuts and dried fruits (unit: 100 gram)			
Budget PL Sales	65.43 (21.74)	17.76 (9.70)	-
Budget PL Profits	59.92 (34.98)	15.38 (10.61)	-
Standard PL Sales	123.24 (39.53)	59.87 (30.06)	-
Standard PL Profits	301.73 (117.76)	147.75 (91.09)	-
Premium PL Sales	50.28 (15.02)	33.26 (11.11)	-
Premium PL Profits	143.53 (41.64)	94.71 (32.14)	-

Notes: Reported across 233,456 shopping trips from 5,384 customers from the same demographic area visiting these store formats. The table presents the mean unit sales (SD) and mean profits (SD) for each PL tier across five categories. The unit of profits is in the local currency.

Table 6: Descriptive statistics of multitier PLs and national brand (NB) pricing across store formats

	Hypermarket	**Supermarket**	**Convenience**
Biscuit (unit: 100 gram)			
Budget PL Price	1.71 (0.13)	1.73 (0.22)	-
Standard PL Price	2.97 (0.34)	3.54 (0.46)	-
NB Price	3.92 (0.43)	4.26 (0.41)	-
Canned Vegetables (unit: 100 gram)			
Budget PL Price	1.71 (0.08)	1.65 (0.11)	1.93 (0.22)
Standard PL Price	2.61 (0.20)	2.70 (0.34)	2.91 (0.56)
Premium PL Price	2.71 (0.43)	2.79 (0.32)	-
NB Price	3.47 (0.26)	3.69 (0.35)	4.36 (1.44)
Cereals (unit: 100 gram)			
Budget PL Price	1.65 (0.08)	1.75 (0.19)	-
Standard PL Price	2.92 (0.15)	3.03 (0.11)	-
Premium PL Price	2.51 (0.25)	-	-
NB Price	3.99 (0.42)	4.37 (0.51)	-
Pasta (unit: 100 gram)			
Budget PL Price	1.22 (0.10)	1.04 (0.12)	-
Standard PL Price	1.47 (0.18)	1.54 (0.25)	-
Premium PL Price	2.46 (0.48)	2.08 (0.52)	-
NB Price	1.48 (0.12)	1.61 (0.12)	-
Raw nuts and dried fruits (unit: 100 gram)			
Budget PL Price	6.75 (0.69)	6.80 (0.65)	-
Standard PL Price	7.67 (0.59)	7.75 (1.18)	-
Premium PL Price	11.51 (1.17)	11.96 (1.59)	-
NB Price	8.20 (1.68)	7.80 (1.76)	-

Notes: Reported across 233,456 shopping trips from 5,384 customers from the same demographic area visiting these store formats. The table presents the mean price per unit (SD) across five categories. The unit of price is in the local currency.

Table 7: Estimation Steps and Methods

Purpose	**Approach**	**Relevant Variables/ Parameter**
Step 1: Estimating Sales-Response Model given interdependent price dynamic among the tiers	Employing Error Correction Model	Change in (quantity) sales as a dependent variable and changes in prices across multi-tier PLs as independent variables
Step 2.1: Calculating elasticities	Normalizing estimated parameters	Price coefficients ($\hat{\alpha}_{1ijk}$, $\hat{\beta}_{(i,l)jk}$) from estimated model in Step 1
Step 2.2: Calculated expected changes in sales given interdependent price dynamics among the tiers	Fitting changes in sales in the estimated model from step 1	Estimated changes in sales for given changes in own prices and changes of other PL tiers prices at t ($\Delta\hat{S}_{ijk,t}$)
Step 3: Estimating Profits-Response Model	Employing Error Correction Model	Change in profits as a dependent variable and changes in estimated sales ($\Delta\hat{S}_{ijk,t}$) from given changes in price across multi-tier PLs at t and its own price change as independent variables
Step 4: Calculating responsiveness of profit to a change in price	Normalizing estimated parameter	Price coefficients ($\hat{\tau}_{1ijk}$) from estimated model in Step 3

format k is:

$$
\begin{aligned}
(1)\quad \Delta S_{ijk,t} = {} & \alpha_0 + \alpha_{1ijk}\Delta Price_{ijk,t} + \sum_{l=1}^{L-1}\beta_{(i,l)jk}\Delta Price_{ljk,t} \\
& + \beta_{2jk}\Delta NBPrice_{jk,t} + \beta_{3ijk}FD_{ijk,t} \\
& + \gamma_{ijk}[S_{ijk,t-1} - \theta_0 - \theta_{1ijk}Price_{ijk,t-1} - \sum_{l=1}^{L-1}\vartheta_{(i,l)jk}Price_{ljk,t-1}] \\
& + \beta_{4ijk}Holiday_t + \beta_{5ijk}Week_t + \beta_{6ijk}Year_t \\
& + \beta_{7ijk}CopulaPrice_{ijk,t} + \beta_{7ijk}CopulaNBPrice_{jk,t} + \varepsilon_{ijk}
\end{aligned}
$$

where $\Delta X_t = X_t - X_{t-1}$ and subscript $(i,l)jk$ denotes competing PL tier of PL tier i within category j in store format k. The sum over l runs from 1 to $L-1$, where L depends on the number of PL tiers available in each category and format. In practice, L could be 2 or 3, as not all categories and store formats offer all three tiers of PLs. In addition to variables of interest, we added the Gaussian copula for $Price_{ijk,t}$ and $NBPrice_{jk,t}$ to address the potential presence of endogeneity (e.g., idiosyncratic changes in consumer preferences for brands) (Datta et al., 2022).

In the specified error-correction model, the coefficient γ_{ijk} captures the speed and extent of adjustment back to the long-term equilibrium of PLs sales following a deviation (Pauwels et al., 2007). This coefficient multiplies the error correction term, which is the difference between the actual sales level of the previous period and its predicted equilibrium level based on their own prices and other competing PL tier prices. This coefficient helps quantify the internal feedback mechanism within the sales process, providing insights into the underlying stability and responsiveness of the market to pricing strategies across PL tiers over time. Moreover, to ensure valid inference given the time-series structure of the data, we correct the estimated standard errors for potential serial correlation and heteroskedasticity using the Newey–West (HAC) estimator (Newey & West, 1987). This correction is applied to all error-correction models used in the analysis.

In this error-correction model, we obtain short-term effects for its own price ($\hat{\alpha}_{1ijk}$) and cross PL tiers' prices ($\hat{\beta}_{(i,l)jk}$) on sales for each competing PL tier. Since different categories have different units of sales, we control for scale differences by converting the obtained parameters ($\hat{\alpha}_{1ijk}$, $\hat{\beta}_{(i,l)jk}$) to unit-free elasticities at mean. This is done by normalizing the estimated parameters with the ratio of the sample mean[3] of prices of each tier or competing tier(s) to sample mean of sales of each tier (Srinivasan et al., 2004; Datta et al., 2022). Thus we get,

$$\hat{\eta}_{1ijk} = \hat{\alpha}_{1ijk} * \frac{\overline{(Price}_{ijk})}{\overline{(Sales}_{ijk})}$$

$$\hat{\eta}_{(i,l)jk} = \hat{\beta}_{(i,l)jk} * \frac{\overline{(Price}_{ijk})}{\overline{(Sales}_{(i,l)jk})}$$

where $\hat{\eta}_{1ijk}$ is the estimated price elasticity for PL tier i and $\hat{\eta}_{(i,l)jk}$ is the estimated cross-price elasticities. We then compare these elasticities across categories and store formats.

Since we want to account for the changes in sales that are not isolated within tiers but are interdependent across the entire range of PLs due to cross-elasticity effects and potential cannibalization within the product line, we calculate estimated changes in sales for given changes in own prices and changes of other PL tiers prices at t, $\Delta\hat{S}_{ijk,t}$, using the estimated model (1). This estimated changes in sales ($\Delta\hat{S}_{ijk,t}$) is used in the subsequent profits-response model.

4.2 Profits-Response Model Specification

In order to quantify the effect of price changes on profits across different PL tiers, we adopt an error correction specification. This specification allows us to capture both short-term effects and long-term adjustments in profitability in response to changes in sales, prices and costs. This is necessary for understanding the immediate impacts as well as the more gradual adjustments that might occur in a grocery retail context. Moreover, it can handle scenarios where variables do not necessarily have to be cointegrated, offering flexibility in our modeling approach. Thus, we can estimate how changes in pricing affect total profits in the short run while also observing how these changes contribute to achieving a long-term equilibrium in profitability. The choice

[3] Since we observe irregular drops of regular prices in certain products, we use the sample median instead of mean to avoid the extreme values in time-series.

of implementing EC specification in the profits-response model also follows logically from its application in the sales-response model, maintaining consistency in our analysis.

Since the profits of PL tier i is influenced by its own pricing adjustments and corresponding changes in sales volumes as proposed, the model for PL tier i profits in category j in store format k is:

$$\begin{aligned}(2)\quad \Delta\pi_{ijk,t} &= \tau_0 + \tau_{1ijk}\Delta Price_{ijk,t} + +\tau_{2ijk}\Delta\hat{S}_{ijk,t} + \tau_{3jk}\Delta cost_{ijk,t} \\ &+ \kappa_{ijk}[\pi_{ijk,t-1} - \rho_0 - \rho_{1ijk}S_{ijk,t-1} - \rho_{2ijk}Price_{ijk,t-1} - \rho_{3ijk}cost_{ijk,t-1}] \\ &+ \tau_{4ijk}Holiday_t + \tau_{5ijk}Week_t + \tau_{6ijk}Year_t \\ &+ \tau_{7ijk}CopulaPrice_{ijk,t} + \upsilon_{ijk}\end{aligned}$$

where $\Delta X_t = X_t - X_{t-1}$ and $\Delta\hat{S}_{ijk,t}$ is obtained from the sales-response models.

In our profits-response model (2), we incorporate the term $\Delta\hat{S}_{ijk,t}$, which represents the estimated changes in sales for a particular PL tier, given the changes in prices of its own and other PL tiers. This variable is derived from the sales-response models and used to quantify how price adjustments in one tier influence the sales volumes and consequently the profits of another tier within the same category and store format. The inclusion of this estimated value allows our model to dynamically capture the ripple effects of price changes across PL tiers, acknowledging that pricing decisions in one tier can have significant spill-over effects on other tiers. We ensure that our profit analysis accounts for these interdependencies relationship and provides a more accurate estimation of profitability impacts, as it considers both direct and indirect effects of price changes. As in the sales-response estimation, we use Newey–West (HAC) standard errors to account for possible serial correlation and heteroskedasticity in the residuals of the profit-response models.

In the specified error-correction model, the coefficient κ_{ijk} captures the speed and extent of adjustment back to the long-term equilibrium of PLs profits following a deviation. It multiplies the error correction term, which is the difference between the actual profits level of the previous period and its predicted equilibrium level based on their own sales, prices and costs. This coefficient helps quantify the internal feedback mechanism within the profits generating process.

Controlling for dynamics of sales and prices across PL tiers, we are interested in short-term effects of changes in price on profits ($\hat{\tau}_{1ijk}$). Since different categories have different profit margin structure, we control for scale differences by converting the this estimated effect of price on profit ($\hat{\tau}_{1ijk}$) to unit-free elasticities at mean. This is done by normalizing the estimated parameter with the ratio of the sample mean of prices to sample mean of profit. Thus we get,

$$\hat{\Theta}_{1ijk} = \hat{\tau}_{1ijk} * \frac{(\overline{Price}_{ijk})}{(\overline{Profits}_{ijk})}$$

5 Results

5.1 Sales-Response Model

We report the results from Sales-Response model in table 8. Across all categories, own price elasticities are significant negative for the standard and premium tiers, indicating that price increases generally lead to reductions in sales volumes in for these PL tiers. Moreover, the premium tier tends to be more sensitive to price compared to the standard tier. The results for the budget tier are mixed, with some elasticities in particular formats showing positive price elasticity. This implies differing consumer perceptions for the budget PL, possibly influenced by perceived value. Cross-elasticities also exhibit mixed effects, suggesting both downward substitution and upward substitution.

In the biscuits category, the significant negative elasticity of the standard tier in both hypermarket and supermarket reflects a sensitivity to price increases, aligning with the general pattern observed across other categories. The budget tier, however, shows a significant positive elasticity in the hypermarket, suggesting that lower prices can drive up sales, potentially indicating a strong consumer preference for value in this category. There is no statistical evidence of substitution between tiers.

In the canned vegetables category, the significant negative elasticity of the standard tier are significantly negative across all three formats. Unlike the biscuits category, the budget tier shows a significant negative elasticity in the supermarket and convenience store suggesting price-conscious consumers for this category. There exhibits upward substitution in the budget tier for this category in the supermarket and convenience store. This mean that an increase in budget PL price in supermarket and convenience store leads to a slight increase in the standard PL sales.

In the cereals category, the premium tier exhibits very high negative elasticity in the hypermarket, suggesting that consumers are extremely price-sensitive and likely to switch to different alternatives when prices rise. Since there is no statistical evidence of substitution among PL tiers, none of the PL tier gains from this.

In the pasta category, the average elasticity is low compared to other categories suggesting the potential necessity of this category. Sales of budget PL do not seem to be significantly influenced by any PL tier price, including its own. However, there is a downward substitution in the supermarket. This mean that increase in premium PL in supermarket leads to an increase in the budget PL sales.

In the raw nuts, dried fruits, and granola bars category, sales of budget PL and premium PL are not significantly influenced by their own prices. There exhibit both upward substitution and downward substitution among PL tiers in both hypermarket and supermarket. For hypermarket, increase in budget PL price leads to an increase in premium PL sales while increase in premium PL price affect negatively to standard PL sales. An increase in standard PL price has a significant

Table 8: Estimated (cross) price elasticity ($\hat{\eta}$) on sales among multitier PLs (B: Budget, S: Standard P: Premium) by category across store formats from Sales-Response model and long-term adjustment coefficient ($\hat{\gamma}_{ijk}$) and adjusted R^2 of the model.

Category	**Format**	$\hat{\eta}$	**Sales B**	**Sales S**	**Sales P**	$\hat{\gamma}_{ijk}$	R^2
Biscuits	Hypermarket	Price B	2.37**	-0.26	NA	-0.70	0.4182
		Price S	0.87	-2.43**	NA	-0.91	0.6295
		Price P	NA	NA	NA	NA	NA
	Supermarket	Price B	-0.26	-0.39	NA	-0.94	0.5194
		Price S	0.65	-4.92***	NA	-0.97	0.7354
		Price P	NA	NA	NA	NA	NA
Canned Vegetables	Hypermarket	Price B	0.15	-0.20	-0.04	-1.0	0.5871
		Price S	-0.24	-2.06***	0.25	-1.0	0.8227
		Price P	-0.05	-0.06	-2.79***	-1.0	0.9428
	Supermarket	Price B	-1.30***	0.46*	0.22	-0.8	0.6212
		Price S	-0.24*	-2.56***	-0.35	-0.8	0.8819
		Price P	-0.21	-0.08*	-0.59	-0.8	0.5987
	Convenience	Price B	-2.32***	0.03***	NA	-0.9	0.7221
		Price S	-0.36	-2.09***	NA	-0.9	0.7703
		Price P	NA	NA	NA	NA	NA
Cereals	Hypermarket	Price B	-2.54***	-0.32	0.34	-1.09	0.6990
		Price S	0.49	-1.87**	-1.01	-1.00	0.5563
		Price P	-0.08	0.03***	-5.99***	-0.98	0.8144
	Supermarket	Price B	-1.36***	-0.13	NA	-1.09	0.5950
		Price S	0.18	-2.76***	NA	-0.83	0.5515
		Price P	NA	NA	NA	NA	NA
Pasta	Hypermarket	Price B	-0.40	0.22	0.55	-1.05	0.5576
		Price S	-0.02	-0.94***	0.41**	-1.03	0.6806
		Price P	0.04**	0.23	-1.81***	-0.85	0.7290
	Supermarket	Price B	0.25	0.33	-1.85	-0.90	0.5154
		Price S	0.14	-0.75***	-0.24	-1.08	0.6487
		Price P	0.25**	0.11	-0.86***	-0.92	0.5926
Raw Nuts, Dried Fruits & Granola Bars	Hypermarket	Price B	0.72	1.19	0.87*	-0.86	0.5397
		Price S	0.65**	-1.15***	-0.97	-0.79	0.4979
		Price P	-1.13	-0.69*	-0.47	-1.13	0.6507
	Supermarket	Price B	1.42*	1.03	-0.69	-0.86	0.5793
		Price S	-1.07*	-0.73	-0.25	-0.88	0.5921
		Price P	0.73*	-0.59	0.37	-1.00	0.6476

Note: *p<0.1; **p<0.05; ***p<0.01

positive effect on budget PL sales in the hypermarket but a negative effect in the supermarket.

The adjustment coefficients ($\hat{\gamma}_{ijk}$) from the Error Correction Model vary significantly across different categories, tiers and formats. The coefficients, being negative, indicate a consistent tendency for prices and sales to revert to equilibrium after deviations, but at different rates. For instance, categories with more negative coefficients, such as cereals, suggest a rapid response to price changes, which could be indicative of high consumer price sensitivity or a competitive market where deviations from typical pricing are quickly corrected. Conversely, categories with less negative coefficients, such as biscuits, might exhibit slower adjustments.

5.2 Profits-Response Model

We report the results from Profits-Response model in table 9. Across all categories, controlling for dynamics of sales and prices across PL tiers, effects of price on profits are significantly positive, as expected, though they vary considerably by category and store format. Given that these elasticities are normalized at the mean and unit-free, it is reasonable to observe higher elasticities for the budget tiers. This is attributed to their lower mean prices, which cause any effect on profit to appear disproportionately high relative to their mean value. Unlike results from the Sales-Response model, the correction coefficients in the Profits-Response model differ, indicating varying speeds at which profit adjustments occur towards equilibrium after price changes. This variation in correction coefficients, particularly the positive effects observed in the standard tier, implies that price adjustments in this tier can lead to a higher equilibrium state of profits.

In the biscuits category, the basic PL tier has higher price elasticity on profits in the hypermarket, suggesting that this tier is relatively more profitable in this format while the standard tier seems to generate more profits in the supermarket. The low adjustment coefficients, especially for basic PL in hypermarket, suggest a gradual return to profit equilibrium, potentially allowing for sustained periods of higher profitability following price adjustments.This combination of high profit elasticity and slow adjustment may be partly explained by our earlier finding from the Sales-Response model, where the basic tier in the hypermarket exhibited a positive relationship between price and sales. In this case, price increases not only boost margins but may also lead to increased demand. This uncommon but strategically valuable scenario helps sustain elevated profit levels over a longer period.

In the canned vegetables category, the supermarket tend to have higher profitability from price changes across PL tiers suggesting a potential prominent presence and perceived values of private labels in the supermarket. This might due to the price elasticity from Sales-Response model of the basic tier that is lower in the supermarket compared to the other tiers, together with significant cross-price elasticity in the supermarket. The convenience store shows relatively lower profit elasticities across tiers. This may be due to a combination of factors: lower reliance on private labels in the convenience format, higher operational costs, and slimmer profit margins. The

Table 9: Estimated price elasticity ($\hat{\Theta}$) on profits for multitier PLs (B: Budget, S: Standard P: Premium) by category across store formats from Profits-Response model and long-term adjustment coefficient ($\hat{\kappa}_{ijk}$) and adjusted R^2 of the model.

Category	**Format**	$\hat{\Theta}$	**Profits**	$\hat{\kappa}_{ijk}$	R^2
Biscuits	Hypermarket	Price B	6.83***	-0.07	0.53
		Price S	3.76***	-1.08	0.51
		Price P	NA	NA	NA
	Supermarket	Price B	5.93***	-0.64	0.51
		Price S	4.30***	-0.51	0.42
		Price P	NA	NA	NA
Canned Vegetables	Hypermarket	Price B	7.45***	-0.64	0.61
		Price S	3.87***	-1.23	0.68
		Price P	3.49***	-1.30	0.63
	Supermarket	Price B	9.10***	-1.14	0.70
		Price S	3.62***	-0.99	0.64
		Price P	4.29***	-1.68	0.65
	Convenience	Price B	5.00***	-1.11	0.62
		Price S	2.72***	-0.89	0.47
		Price P	NA	NA	NA
Cereals	Hypermarket	Price B	28.56***	-1.16	0.93
		Price S	7.65***	-0.57	0.66
		Price P	6.92***	-0.67	0.59
	Supermarket	Price B	39.65***	-1.17	0.88
		Price S	4.11**	0.15	0.46
		Price P	NA	NA	NA
Pasta	Hypermarket	Price B	6.10***	-1.19	0.71
		Price S	10.04***	-0.66	0.82
		Price P	3.86***	-0.63	0.70
	Supermarket	Price B	5.60***	-1.68	0.70
		Price S	7.25***	-1.21	0.75
		Price P	3.67***	-1.14	0.63
Raw Nuts, Dried Fruits & Granola Bars	Hypermarket	Price B	8.74***	-0.75	0.61
		Price S	4.51***	0.63	0.54
		Price P	4.98***	-1.04	0.67
	Supermarket	Price B	10.07***	-0.91	0.57
		Price S	4.86***	-0.54	0.58
		Price P	5.72***	-1.75	0.68

Note: *p<0.1; **p<0.05; ***p<0.01

adjustment coefficients are moderately negative across all tiers and formats, suggesting a steady but not immediate return to profit equilibrium following price changes. The premium tier exhibits a faster adjustment process. One possible reason is that the premium tier does not get affected from other tiers' prices changes, as shown in the Sales-Response model. This independence may lead consumers to respond more immediately to pricing changes, allowing the tier's profits to stabilize more quickly.

In the cereals category, the basic PL tier appears to be more profitable in the supermarket, while the standard tier shows stronger profit elasticity in the hypermarket. This pattern may be explained by the lower price elasticity of the basic tier in the supermarket and of the standard tier in the hypermarket, as shown in the Sales-Response model. When demand is less sensitive to price increases, retailers can raise prices without significantly reducing sales volume, making such adjustments more effective in increasing overall profits. The adjustment coefficients are generally moderately negative, suggesting that profit levels tend to revert steadily to equilibrium following a price change. However, an exception is observed in the standard tier in the supermarket, which shows a positive adjustment coefficient. This implies that deviations from the profit equilibrium, such as a temporary increase in profits due to a price change, do not necessarily return to a long-term equilibrium. One possible interpretation is that consumers in the supermarket setting may gradually adapt to the new price as a norm, especially for mid-tier products like standard PLs that are positioned as reliable and consistent. This slow or reinforcing response may reflect changes in consumer expectations or perception of value, where the new price point becomes accepted as the standard over time, preventing a reversion to previous profit levels.

In the pasta category, the hypermarket tends to have higher profitability from price changes across PL tiers suggesting a stronger consumer loyalty toward PLs. This is supported by descriptive statistics (Table 6) showing that the average prices of pasta across all PL tiers are higher in the hypermarket than in the supermarket, potentially signaling higher perceived quality or broader product selection in this format. Despite the slightly higher price elasticity across tiers in the hypermarket from the Sales-Response model, the magnitude of profit elasticities suggests that price increases still effectively drive profitability. This indicates that while consumers do react to price changes, the combination of volume, higher margins, and perceived value more than compensates, resulting in overall profit growth. In terms of adjustment, the hypermarket shows a generally slower return to profit equilibrium compared to the supermarket. This slower adjustment may be attributed to more stable or habitual purchasing behavior in the hypermarket. Such patterns may delay immediate responses to price changes, allowing elevated profit levels to persist over longer periods.

In the raw nuts, dried fruits, and granola bars category, the basic tier appears to be more profitable in the hypermarket, while the standard tier and premium tier show stronger elasticity in the supermarket. This finding may be partly explained by the findings from the Sales-Response model: in the hypermarket, the basic tier was less affected by its own price changes, suggesting that modest price increases can improve margins without significantly reducing sales. In contrast, both the standard and the premium tiers in the supermarket showed weaker responses to their own price and minimal cross-price effects, indicating that these products can maintain steady sales even when priced higher, resulting in improved profitability. In terms of adjustment, the premium tier shows a relatively faster return to profit equilibrium in both hypermarket and supermarket. On the other hand, we observe a positive adjustment coefficient for the standard tier in the hypermarket. This suggests that deviations from the standard tier's profit equilibrium may persist or even escalate over time, possibly due to shifting perceptions of value or changes in competitive dynamics within this mid-range offering.

6 Discussion

6.1 How Multitier PLs Perform Across Store Formats?

This study explores how price effectiveness on sales varies across multi-tier private labels (PLs) and store formats, and whether these differences translate into divergent profitability outcomes. Specifically, we examine whether PL tiers, which are budget, standard, and premium, respond differently to pricing strategies depending on the store format (hypermarket, supermarket, or convenience store) within the same retail chain. Using a stepwise empirical framework that links price changes to sales and then to profits, we estimate both immediate and sequential effects of pricing across tiers and formats using Error Correction Models (ECM).

Our findings indicate significant variation in how PL tiers respond to price changes across formats. Across most categories and formats, standard and premium tiers consistently exhibit negative own-price elasticities, confirming that price increases tend to reduce sales volumes. This is consistent with prior research highlighting consumers' price sensitivity for higher-tier PLs (e.g., Gielens, 2012; Geyskens et al., 2010). However, the magnitude of these elasticities differs by format, suggesting that the same PL tier may exhibit stronger or weaker price responsiveness depending on the store format. This supports previous findings that store format plays a moderating role in influencing customer price responses, possibly due to differences in shopper missions, product assortment, or competitive benchmarks (Van der Plas et al., 2024).

In contrast, the budget tier shows greater heterogeneity, with occasional positive elasticities observed, particularly in the hypermarket. These findings align with prior work on price-quality signaling (Rao & Monroe, 1989), suggesting that some consumers may interpret price increases in lower-tier products as an indicator of improved value. These findings highlight the need for retailers to account for consumer perception dynamics when implementing pricing strategies for lower-margin tiers.

Our study shows that cross-tier substitution is limited across most formats and categories. Premium PLs show minimal responsiveness to price changes in lower tiers, implying that consumers view them as distinct rather than interchangeable. This is consistent with literature on niche differentiation and quality segmentation (Martos-Partal et al., 2013). This weak cross-tier substitution suggests that multi-tier PLs can often function as independent offerings rather than internal substitutes, giving retailers more flexibility to pursue differentiated pricing strategies without triggering extensive cannibalization.

While substitution across PL tiers is limited, the observed changes in sales indicate that demand adjustments are likely driven by sources beyond the PL portfolio itself. At the same time, the limited and mixed substitution patterns within PL tiers suggest that cross-tier responses depend on their relative positioning, particularly the price gap between adjacent tiers, where wider gaps may encourage downward substitution and narrower gaps may facilitate upward movement. Given the scope of our data, which is limited to selected stores within a single retail chain, we cannot directly identify whether these changes reflect switching from national brands, shifts across competing retailers, or changes in overall consumption levels. However, several plausible mechanisms may explain these patterns. First, private labels often compete more directly with national brands than with other PL tiers, suggesting that price changes may primarily shift demand between PLs and national brands rather than across PL tiers. Second, format-specific shopping missions may influence how demand responds; for example, in hypermarkets, price changes may affect basket size or category-level purchases, while in convenience stores, purchases are more mission-driven and less sensitive to price changes. Finally, the limited substitution across tiers may reflect relatively stable preferences within each tier, where consumers adjust quantities rather than switch between tiers in response to price changes.

These differentiated sales responses carry through to profitability. Results from the Profits-Response models indicate that pricing adjustments have a positive impact on profits across tiers, although the extent of this impact varies. Budget tiers, despite lower margins, tend to show stronger profit responsiveness, especially in hypermarket format since small price changes can yield relatively larger sales and margin gains. This aligns with recent work by Keller, Dekimpe, and Geyskens (2022), who emphasized the disproportionate profit impact from pricing actions in lower-tier PLs. Meanwhile, standard and premium tiers, although typically less elastic, still generate profitability gains, especially in store formats where consumers recognize their added value and are less likely to switch to lower-tier alternatives.

The comparative findings across formats suggest that PL performance is format-contingent. Hypermarkets benefit most from pricing actions on budget PLs, while supermarkets derive more profit gains from standard and premium PLs. In contrast, the convenience store, which is characterized by its limited assortment and simplified product mix, shows more muted responsiveness to pricing, likely due to fewer options available for substitution and a focus on immediacy and routine purchases rather than price-driven shopping.

Taken together, our findings suggest that while retailers already differentiate pricing across store formats, these strategies can be refined using evidence on tier-specific responsiveness and profitability. Rather than adjusting prices by rule of thumb or benchmarking across formats, retailers should base price decisions on the elasticity–profitability patterns identified for each tier. For example, in hypermarkets, where price sensitivity in budget tiers may be lower or even positive, retailers could explore selective price increases on budget PLs to improve margins without harming volume. In supermarkets, where standard and premium PLs show stronger price sensitivity but also yield higher profitability, moderate, carefully timed adjustments can help increase margins while maintaining quantity sold. In convenience stores, with limited assortment and higher price sensitivity, it may be most effective to maintain stable pricing on core products to retain baseline volumes and avoid deterring quick-stop shoppers. Overall, pricing refinement should rely on empirical monitoring of tier-specific elasticities rather than uniform or intuition-based markups.

Throughout this study, we present our results separately by category, tier, and store format to capture the heterogeneity in pricing responses across these dimensions. Given the limited number of product categories and the uneven availability of tiers across formats in our setting, aggregating these effects into a single summary measure would mask meaningful differences in how pricing operates across tiers and formats. This level of disaggregation also allows for clearer interpretation within each category, without conflating effects across different units of analysis, even though the discussion synthesizes the findings at a higher level.

While these findings highlight the role of pricing in shaping format-specific performance, price is only one element of the broader retail marketing mix (Blut et al., 2018; Gielens & Steenkamp, 2025). Retailers also differentiate store formats through assortment, promotions, and store characteristics, which can influence consumer behavior both independently and through their interactions with one another, including but not limited to pricing decisions. Our analysis isolates the price–sales–profit relationship to provide a clearer understanding of pricing effectiveness, but this should not be interpreted as suggesting that price is the dominant lever for format differentiation. Rather, pricing should be considered alongside other strategic elements that jointly influence consumer behavior across formats. Finally, our conclusions are derived from a retailer-focused perspective based on sales and profitability outcomes. While our framework captures consumer responses through observed purchasing behavior, it does not directly measure consumer perceptions such as price fairness or brand image.

6.2 Limitations

This study has several limitations, each of which presents opportunities for future research. First, our empirical results are inherently dependent on the nature and scope of our dataset, which primarily consists of grocery retailer data collected from consumers whose initial purchases were made at the hypermarket in a certain area. Consequently, our findings may not capture the full spectrum of shopping behaviors or preferences evident among the broader population. This limits the generalizability of our results to other customer segments and retail contexts. Future research could address this limitation by analyzing more diverse datasets, including consumers from multiple retail formats or less brand-loyal segments, to verify the robustness and general applicability of our conclusions.

Second, our results may be sensitive to the data selection criteria applied to ensure comparability across store formats and product tiers. In constructing the dataset, we restrict the sample to customers whose initial purchase occurred in the hypermarket, focus on a limited number of stores within a single city, and include only frequently purchased products and categories that are consistently available across formats. While these criteria help reduce noise from sparse observations and avoid confounding effects from differences in product availability, they may also limit the representativeness of the sample. As a result, the findings should be interpreted within the context of a controlled empirical setting rather than as fully generalizable across all customers, stores, and product categories. Future research could examine the robustness of these results by relaxing these selection criteria or using broader datasets that capture more diverse shopping patterns.

Third, due to data limitations, we did not incorporate additional marketing mix elements, such as advertising and promotional activities, which are known to substantially influence consumer decision-making processes. Omitting these variables does not invalidate our results, as the analysis isolates price–sales–profit dynamics within consistent store and category settings. However, it could potentially limit the completeness of interpretation, as unobserved promotional effects might amplify or attenuate measured price responses (Bijmolt et al., 2005). Future studies could thus enhance the comprehensiveness of results by integrating detailed data on marketing communications, promotional strategies, and in-store displays, enabling a deeper understanding of how these factors interact with pricing decisions to shape consumer behaviors and retail outcomes. Moreover, the use of retailer-level transactional data limits our ability to directly observe consumer perceptions, such as price fairness or brand image. As a result, our findings reflect the retailer side of pricing outcomes rather than both the retailer and consumer perspectives.

Fourth our analysis assumes interdependence among multi-tier PLs to estimate potential cross-price effects, yet the observed limited cross-tier elasticities in certain categories suggest that consumers may perceive these tiers as distinct and non-substitutable. This limitation implies that the assumed inter-tier relationships might not reflect actual consumer perceptions or behaviors across all product categories. Future research might employ consumer perception surveys or discrete-choice models to explicitly test and confirm the degree of substitution between PL tiers and refine the understanding of these relationships across different product categories and market segments (Gielens et al., 2021).

Fifth, while we explicitly model differentiation across private label tiers, we do not distinguish between tiers of national brands due to the absence of consistent tier classifications in the data. While price differences exist across national brands, these do not map cleanly into stable tier structures across categories. As a result, our analysis captures the overall competitive price environment rather than more granular brand-level positioning. Future research could explore more detailed competitive structures by incorporating richer brand-level information or external classification schemes.

Finally, our use of linear Error Correction Models (ECM) presupposes specific structural relationships between price, cost, sales, and their lagged effects, potentially restricting our analysis to linear dynamics. The reality of consumer behavior and pricing responses may involve more complex, nonlinear interactions not fully captured by our modeling approach. Future research could extend our findings by exploring alternative modeling frameworks, such as nonlinear or machine-learning-based approaches, that might capture more nuanced and dynamic relationships between prices, sales, and profitability.

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